EXERCISE#01

Editor - C:\Users\hasee\OneDrive\Documents\MATLAB\DSPLAB5\DTFT.m

DTFT.m | EXERCISE2.m | EXERCISE3.m | EXERCISE4.m | E

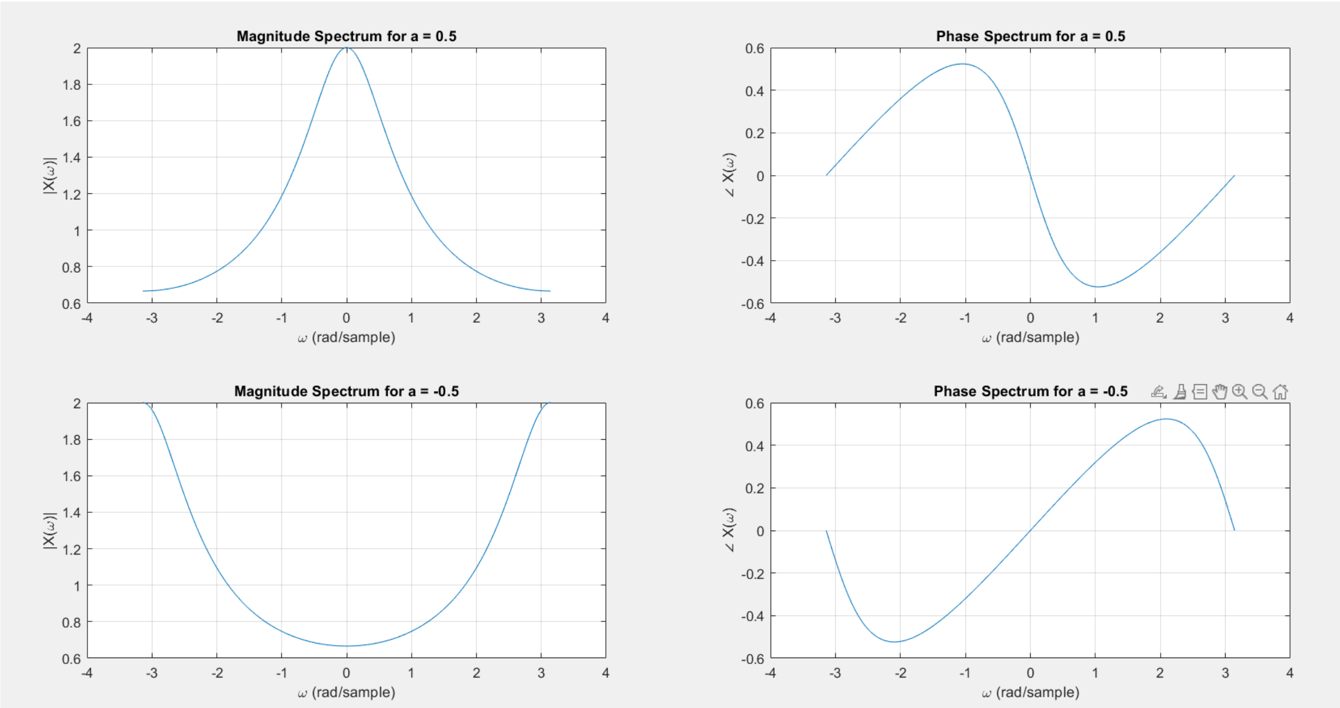
```matlab
function y = DTFT(x, n, w)
    y = zeros(size(w));
    for k = 1:length(w)
        y(k) = sum(x .* exp(-1j * w(k) * n));
    end
end
```

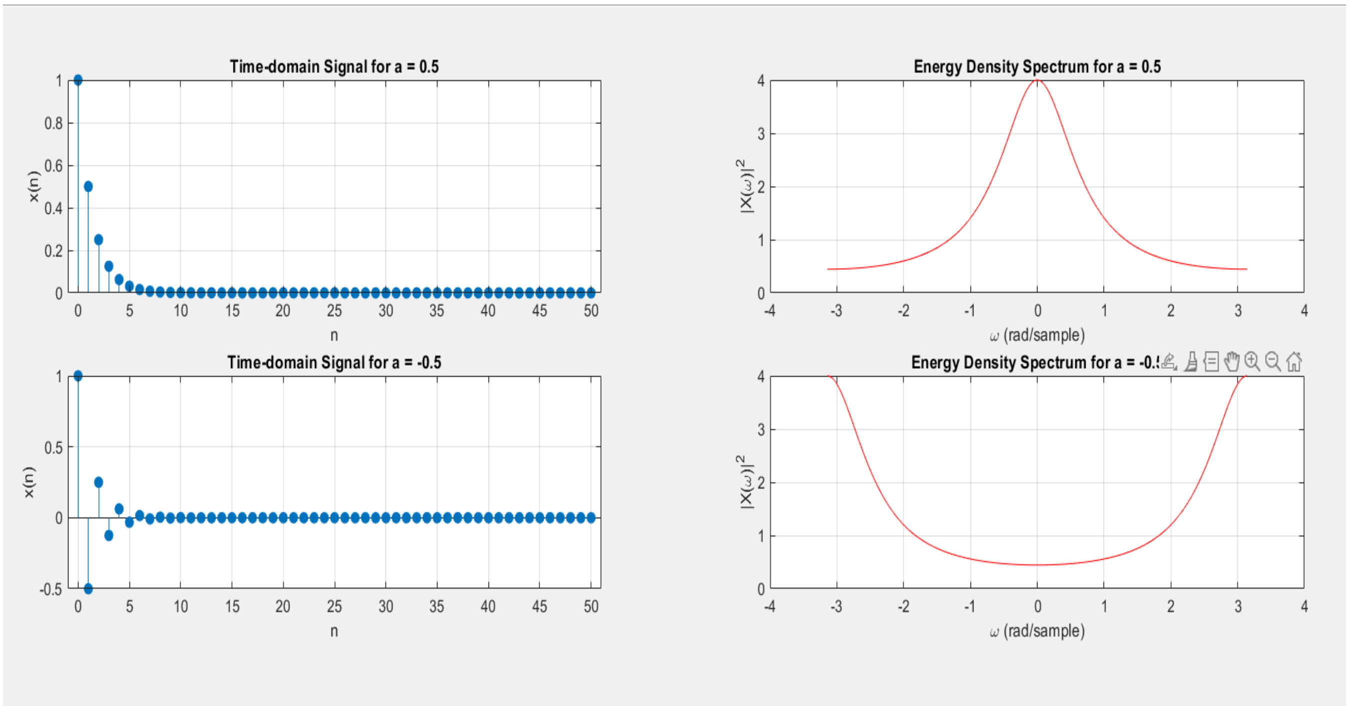
EXERCISE#02

Editor - C:\Users\hasee\OneDrive\Documents\MATLAB\DSPLAB5\EXERCIS

DTFT.m | EXERCISE2.m | EXERCISE3.m | EXERCISE4.m

```matlab
n = 0:50;
w = linspace(-pi, pi, 500);
a_values = [0.5, -0.5];

figure;
for i = 1:length(a_values)
    a = a_values(i);
    x = a.^n;
    Xw = DTFT(x, n, w);
    Sxx = abs(Xw).^2;

    subplot(3,2,2*i-1);
    stem(n, x, 'filled');
    xlabel('n'); ylabel('x(n)');
    title(['Time-domain Signal for a = ', num2str(a)]);
    grid on;

    subplot(3,2,2*i);
    plot(w, Sxx, 'r');
    xlabel('\omega (rad/sample)');
    ylabel('|X(\omega)|^2');
    title(['Energy Density Spectrum for a = ', num2str(a)]);
    grid on;
end
figure;
for i = 1:length(a_values)
    a = a_values(i);
    x = a.^n;
    Xw = DTFT(x, n, w);


    subplot(2,2,2*i-1);
    plot(w, abs(Xw));
    xlabel('\omega (rad/sample)');
    ylabel('|X(\omega)|');
    title(['Magnitude Spectrum for a = ', num2str(a)]);
    grid on;

    subplot(2,2,2*i);
    plot(w, angle(Xw));
    xlabel('\omega (rad/sample)');
    ylabel('\angle X(\omega)');
    title(['Phase Spectrum for a = ', num2str(a)]);
    grid on;
end
```


Magnitude Spectrum for a = 0.5
Phase Spectrum for a = 0.5
Magnitude Spectrum for a = -0.5
Phase Spectrum for a = -0.5
|X(ω)|
∠ X(ω)
ω (rad/sample)

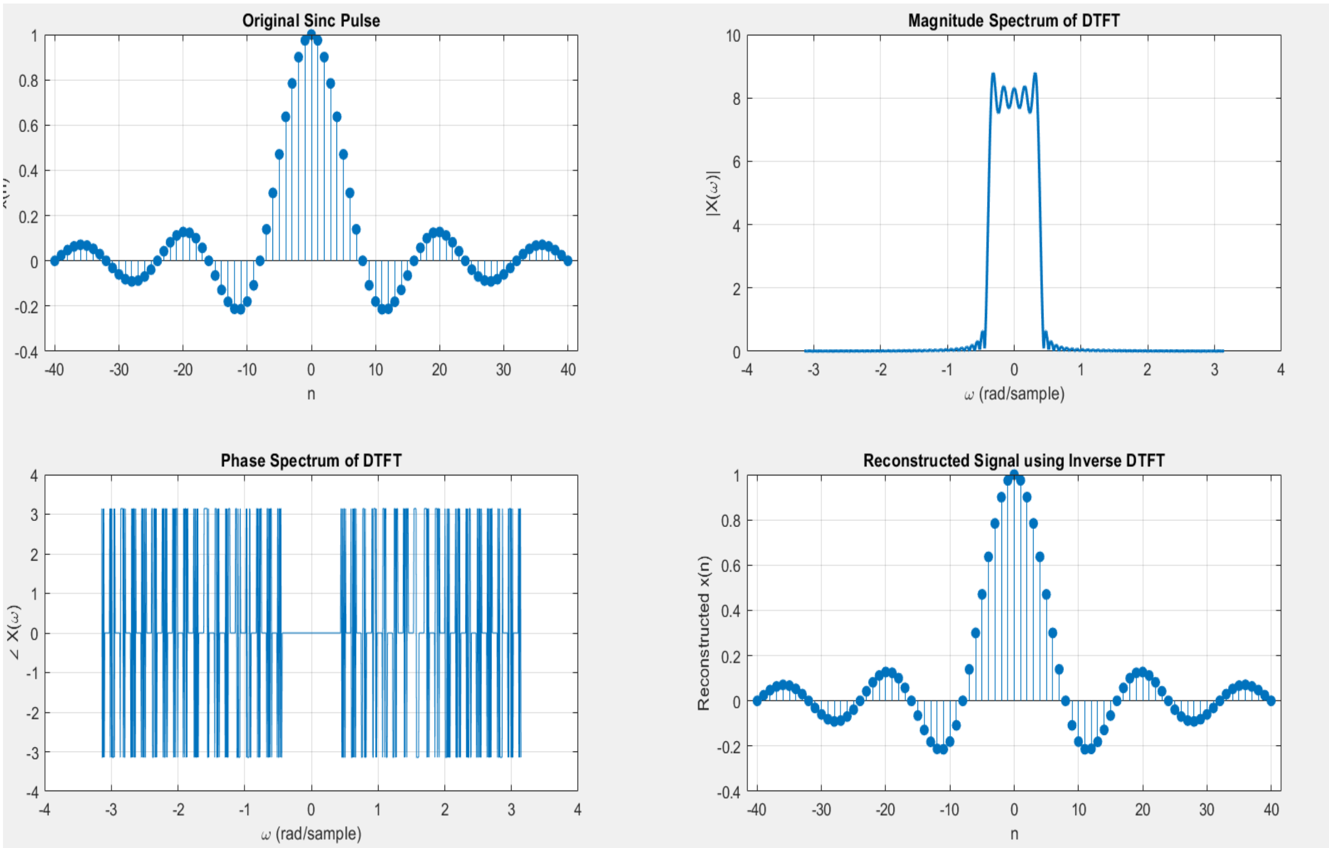

Time-domain Signal for a = 0.5
Energy Density Spectrum for a = 0.5
Time-domain Signal for a = -0.5
Energy Density Spectrum for a = -0.5
x(n)
n
|X(ω)|^2
ω (rad/sample)

EXERCISE#03

Editor - C:\Users\hasee\OneDrive\Documents\MATLAB\DSPLAB5\EXERCISE3.m

DTFT.m | EXERCISE2.m | EXERCISE3.m | EXERCISE4.m | EXERCISE6.m

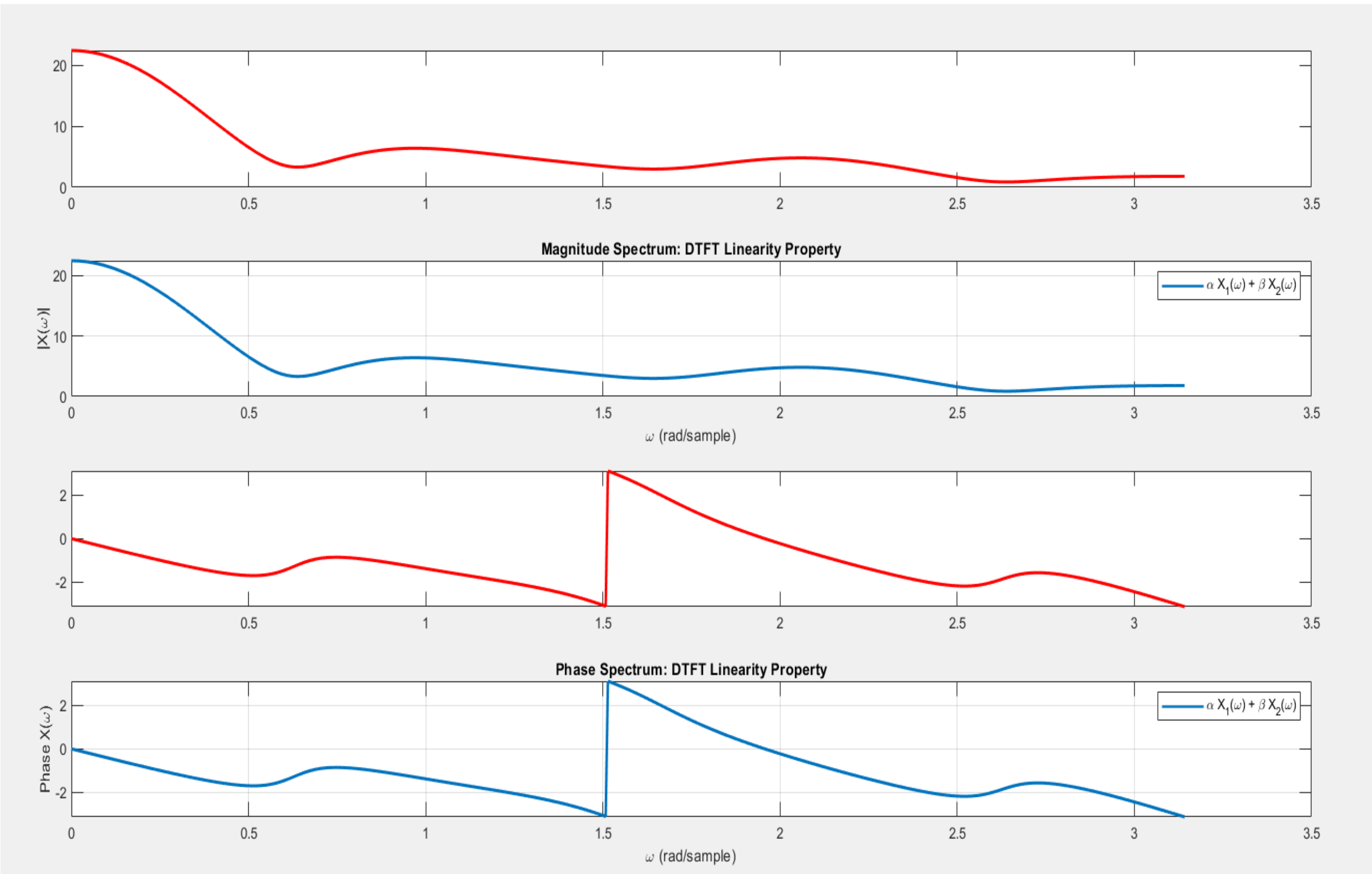
```matlab
n = -40:40;
f = 1/8;
x = sinc(f * n);

w = linspace(-pi, pi, 1000);
Xw = zeros(size(w));

for k = 1:length(w)
    Xw(k) = sum(x .* exp(-1j * w(k) * n));
end

[W, N] = meshgrid(w, n);
dw = w(2) - w(1);
x_reconstructed = real(trapz(w, (1/(2*pi)) * (Xw .* exp(1j * W .* N)), 2));
figure;

subplot(2,2,1);
stem(n, x, 'filled');
xlabel('n'); ylabel('x(n)');
title('Original Sinc Pulse');
grid on;


subplot(2,2,2);
plot(w, abs(Xw), 'LineWidth', 1.5);
xlabel('\omega (rad/sample)'); ylabel('|X(\omega)|');
title('Magnitude Spectrum of DTFT');
grid on;

subplot(2,2,3);
plot(w, angle(Xw), 'LineWidth', 0.5);
xlabel('\omega (rad/sample)'); ylabel('\angle X(\omega)');
title('Phase Spectrum of DTFT');
grid on;


subplot(2,2,4);
stem(n, x_reconstructed, 'filled');
xlabel('n'); ylabel('Reconstructed x(n)');
title('Reconstructed Signal using Inverse DTFT');
grid on;

```



EXERCISE#04

DTFT.m | EXERCISE2.m | EXERCISE3.m | EXERCISE4.m | EXERCISE6.m | EXERCISE7.m | POSTLAB.m

```
plot(w, abs(X_sum_time), 'LineWidth', 1.5);
xlabel('\omega (rad/sample)'); ylabel('|X(\omega)|');
title('Magnitude Spectrum: DTFT Linearity Property');
legend('\alpha X_1(\omega) + \beta X_2(\omega)', 'DTFT of (\alpha x_1(n) + \beta x_2(n))');
grid on;
subplot(4,1,3);
plot(w, angle(X_sum_freq), 'r', 'LineWidth', 1.5); hold on;
subplot(4,1,4);
plot(w, angle(X_sum_time), 'LineWidth', 1.5);
xlabel('\omega (rad/sample)'); ylabel('Phase X(\omega)');
title('Phase Spectrum: DTFT Linearity Property');
legend('\alpha X_1(\omega) + \beta X_2(\omega)', 'DTFT of (\alpha x_1(n) + \beta x_2(n))');
grid on;


function Xw = dtft(x, n, w)
    Xw = zeros(size(w));
    for k = 1:length(w)
        Xw(k) = sum(x .* exp(-1j * w(k) * n));
    end
end

```



EXERCISE#05

ocuments ▸ MATLAB ▸ DSPLAB5

Editor - C:\Users\hasee\OneDrive\Documents\MATLAB\DSPLAB5\EXERCISE5.m

DTFT.m | EXERCISE2.m | EXERCISE3.m | EXERCISE4.m | EXERCISE5.m | EXERCISE7.m | POSTLAB.m | +

```matlab
x = [1 1 1];
n = -1:1;

w = 0:pi/100:2*pi;

Xw = dtft(x, n, w);

E_time = sum(abs(x).^2);

E_freq = trapz(w, abs(Xw).^2) / (2 * pi);

fprintf('Energy computed from time domain: %.4f\n', E_time);
fprintf('Energy computed from frequency domain: %.4f\n', E_freq);


function Xw = dtft(x, n, w)
    Xw = zeros(size(w));
    for k = 1:length(w)
        Xw(k) = sum(x .* exp(-1j * w(k) * n));
    end
end

```

Command Window

New to MATLAB? See resources for Getting Started.

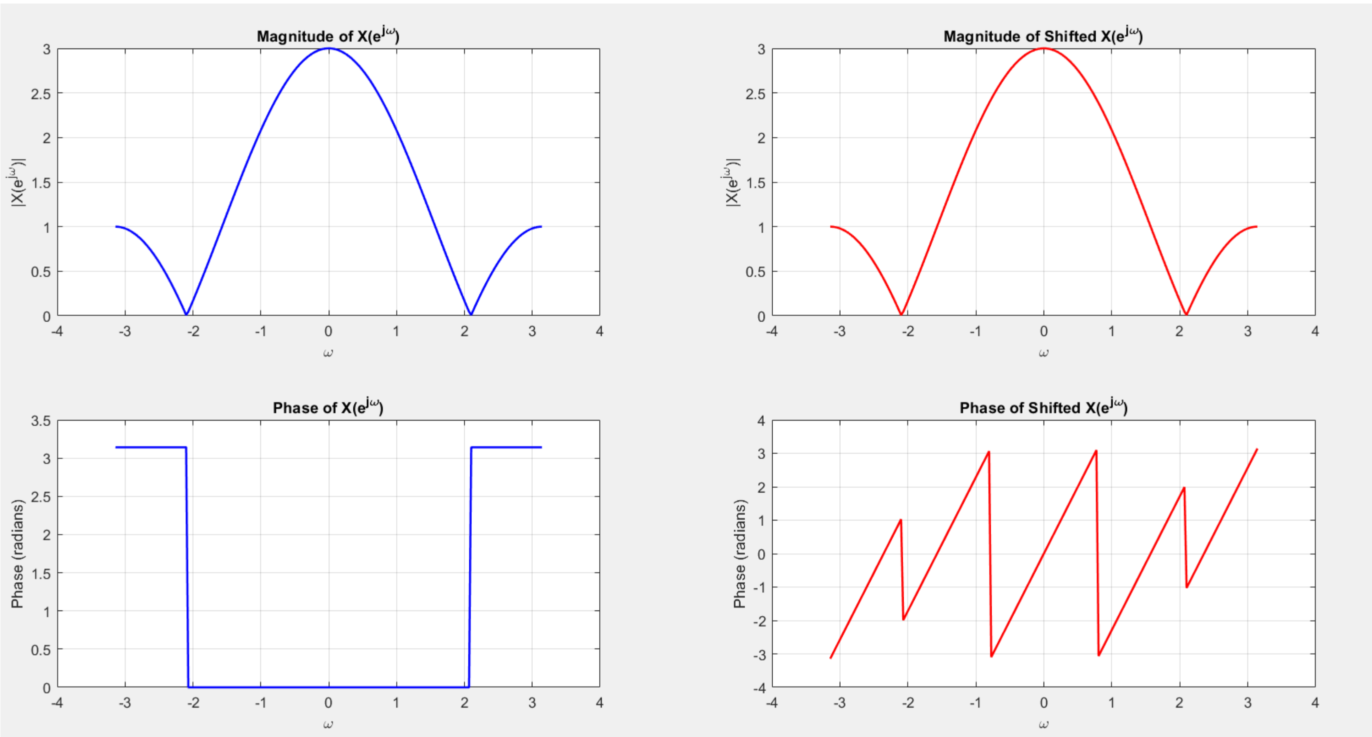
```
>> EXERCISE5
Energy computed from time domain: 3.0000
Energy computed from frequency domain: 3.0000
>>
```

EXERCISE#06

Editor - C:\Users\hasee\OneDrive\Documents\MATLAB\DSP

DTFT.m | EXERCISE2.m | EXERCISE3.m | EX

```matlab
x = [1 1 1];
n = -1:1;

w = linspace(-pi, pi, 200);

Xw = dtft(x, n, w);
d = -4;
n_shifted = n + d;
Xw_shifted = Xw .* exp(-1j * w * d);

figure;
subplot(2,2,1);
plot(w, abs(Xw), 'b', 'LineWidth', 1.5);
title('Magnitude of X(e^{j\omega})');
xlabel('\omega');
ylabel('|X(e^{j\omega})|');
grid on;

subplot(2,2,2);
plot(w, abs(Xw_shifted), 'r', 'LineWidth', 1.5);
title('Magnitude of Shifted X(e^{j\omega})');
xlabel('\omega');
ylabel('|X(e^{j\omega})|');
grid on;

subplot(2,2,3);
plot(w, angle(Xw), 'b', 'LineWidth', 1.5);
title('Phase of X(e^{j\omega})');
xlabel('\omega');
ylabel('Phase (radians)');
grid on;

subplot(2,2,4);
plot(w, angle(Xw_shifted), 'r', 'LineWidth', 1.5);
title('Phase of Shifted X(e^{j\omega})');
xlabel('\omega');
ylabel('Phase (radians)');
grid on;
function Xw = dtft(x, n, w)
    Xw = zeros(size(w));
    for k = 1:length(w)
        Xw(k) = sum(x .* exp(-1j * w(k) * n));
    end
end
```



POST LAB:

Editor - C:\Users\hasee\OneDrive\Documents\MATLAB\DSPL

DTFT.m | EXERCISE2.m | EXERCISE3.m | EXER

```matlab
x = [1/3, 1/3, 1/3];
n = [0 1 2];
w = linspace(-pi, pi, 200);
Xw = dtft(x, n, w);

x_conj = conj(x);
Xw_conj = dtft(x_conj, n, w);

Xw_neg = conj(flip(Xw));

figure;
subplot(2,2,1);
plot(w, abs(Xw), 'b', 'LineWidth', 1.5);
title('|X(e^{j\omega})|');
xlabel('\omega');
ylabel('Magnitude');
grid on;

subplot(2,2,2);
plot(w, abs(Xw_conj), 'r', 'LineWidth', 1.5);
title('|X^*(e^{j\omega})|');
xlabel('\omega');
ylabel('Magnitude');
grid on;

subplot(2,2,3);
plot(w, angle(Xw), 'b', 'LineWidth', 1.5);
title('Phase of X(e^{j\omega})');
xlabel('\omega');
ylabel('Phase (radians)');
grid on;

subplot(2,2,4);
plot(w, angle(Xw_neg), 'r', 'LineWidth', 1.5);
title('Phase of X^*(-\omega)');
xlabel('\omega');
ylabel('Phase (radians)');
grid on;

function Xw = dtft(x, n, w)
    Xw = zeros(size(w));
    for k = 1:length(w)
        Xw(k) = sum(x .* exp(-1j * w(k) * n));
    end
end
```